1. **Introduction:**
   Transfusion in sickle cell disease (SCD) can be lifesaving and reduces morbidity. However, paucity of randomized controlled clinical trials has resulted in wide variations in clinical practice. The most important transfusion related adverse events in patients on chronic transfusion is iron overload and alloimmunization. The transfusion plan must consider the target haemoglobin concentration (Hb) and/or percentage of sickle haemoglobin (%HbS) to ensure maximal oxygen delivery to tissues without increasing overall blood viscosity. During an exchange procedure, the system removes defective red cells (RBC) from the patient and replaces them with healthy donor RBC. During an exchange procedure, RBC are removed until the target fraction of cells remaining (FCR) or the target for replaced volume has been attained.
2. **Scope:**
   This document applies to all adult patients with SCD, who needs red cell exchange at the Royal Hospital.
3. **Responsibilities:**
   Trained staff nurse working in the apheresis unit and trained physicians on the management of patients with SCD.
4. **Policy:**

4.1 <u>Pre-first red cell exchange (RCE):</u>

- All patients must have the following baseline investigations: full blood count (CBC), reticulocyte count, liver function, renal function, HPLC and blood group with antibody screen.
- All SCD patients shall be screened for hepatitis B (HBV), hepatitis C (HCV) and human immunodeficiency virus (HIV).
- Patients with SCD must also have baseline extended RBC antigen typing.
- HBV vaccination should be given to all patients with SCD, irrespective of previous or prospective planned transfusions.
- Consent for blood transfusion should be taken for all patients and renewed annually.
- Targets of red cell exchange shall be defined for each patient and maintained in the medical record.
- Premedication should be given only to patients who had a documented previous allergic reaction secondary to transfusion (Premedication: Paracetamol 1 gm IV/PO, hydrocortisone 100 mg IV, chlorpheniramine 4 mg PO or 10 mg IV)

4.2 <u>Follow up:</u>

- SCD patients on regular transfusions should be screened annually for HBV, HCV, and HIV.
- Chronically transfused SCD patients should be regularly monitored for iron overload with serum ferritin at least every 3 months. Low ferritin levels (1500 ng/mL) generally correlate with well controlled liver iron concentration.
- Liver iron measurements is recommended for patients with suspected or documented transfusional iron overload; should be performed every 1–2 years, using validated non-invasive magnetic resonance imaging techniques.

- Patients on iron chelation therapy, further testing (LFT, UE1) to monitor side-effects should be performed.

4.3 <u>Red cell units' specifications:</u>

- All SCD patients planned for red cell exchange shall receive Rh and K antigen matched red cell units.
- If there are clinically significant red cell antibodies (current or historical) then the red cells selected should be negative for the corresponding antigens
- Red cells units shall be 14 days or less old (but older blood may be given if the presence of red cell antibodies makes the provision of blood difficult).
- Blood provided for SCD patients should be HbS negative and, where possible leucoreduced.

5. Adverse events related to the procedure:

5.1 Citrate toxicity leading to hypocalcemia increases with increasing number of units of blood transfused (refer to HAEM/AU/2/Ver.1.0) for management of hypocalcemia.

5.2 Fluid shifts: hydration status should be addressed pre-procedure; anti-hypertensive medications and diuretics may need to be withheld. Apheresis machines pool a fixed volume of blood ex vivo and where this is >15% of the patient's total blood volume, priming the machine with donor blood prior to apheresis (e.g. patients who have Hb >20% below their steady state Hb) or simple transfusion could be given prior to apheresis.

6. Replacement volume and duration of RCE procedure:

For an effective red cell exchange starting with an untransfused patient, plan on 1.5 red cell volumes administered during the exchange. This is because some of the administered red cells will be removed later in the procedure as the exchange continues. Once a patient is effectively exchanged, chronic exchange can usually be done every 4 weeks with 1 red cell volume with good results for most patients. Some patients will require adjustment to a larger volume (1 or 2 extra units but generally not more than 1.5 red cell volumes), as the post- HbA level will be too low. Some will require a shorter time interval since their lifespan of transfused cells seems shorter.

Estimated total blood volume (TBV) for adult per Kg body weight by sex:
Male: 70 ml/Kg
Female: 65 ml/Kg
Red cell volume (RCV): Haematocrit (Hct) X TBV

<u>Example (can be used when performing manual RCE):</u>

For a 70 kg male with Hct 0.35 undergoing manual 1.5 RCV red cell exchange for acute chest syndrome:
TBV= 70 ml/kg X 70 Kg= 4,900 ml
RCV= 0.35X 4,900= 1,715 ml or 1.5 RCV =1.5X1,715 ml= 2,572.5 ml
(If average packed red cell unit volume is 250 ml and 100% volume replacement): 2,572.5 ml/250 ml=10.3 units

7. Venous access:

- Suitable options for vascular access include:
- Peripheral veins (16–18 gauge preferred)

- Double-lumen central venous catheter (CVC) designed for apheresis or hemodialysis.
- Subcutaneous implantable access device (e.g., Vortex port-a-cath)
- Arteriovenous fistula or graft.

Access through peripheral veins is the modality of choice in most cases because it is associated with fewer infectious, thrombotic, and insertion-related complications (e.g., discomfort, scarring, bleeding or pneumothorax) as compared to central lines. if the draw port is distal to the return port either by confusion of the ports on a single apheresis catheter or if blood is returned to a connected vein on the same arm. The draw site generally needs at least a 19G needle, while the return can be as small as a 21G, so it may be tempting to place a draw line higher up in the arm (i.e., near the antecubital area) and to use a distal vein near the hand for the return. In general, opposite arms should be used for draw and return.

8. Indications for regular RCE:

In general, the target for RCE in SCD as follows:

- End HCT at 30 ± 3% (<33%-36% to avoid hyperviscosity) & HbS (or HbS + variant) of <30%.
- Providers tend to target a lower HbS goal of <15% so that the next pre procedure HbS is <30%.

Indications	Targets
Primary stroke prevention	Target HBS ≤30% and post Hb level >9 gm/dl Transfusions may be discontinued after 2 years if repeat vascular imaging is normal at that time
Secondary stroke prevention	Target HBS ≤30% and post Hb level >9 gm/dl Can raise target HbS level from <30% to <50%, when stroke patients remain neurologically stable for 4 years following the initial event (risk of recurrent stroke is highest in the first 3 years after the initial event)
Recurrent chest syndrome (ACS) with hypoxemia despite optimal Hydroxyurea therapy or intolerance & side effects	Target HBS ≤30% and post Hb level >9 gm/dl
Pregnancy	Can be considered for women with: • Previous or current medical, obstetric or fetal problems related to SCD • Women previously on hydroxyurea because of severe disease • multiple pregnancy There is no evidence to support a specific goal but target Hb level > 7.0 gm/dL & a peak HbS% of <50% pre RCE.
Where transfusion is considered for indications where there is insufficient evidence for its benefit (e.g. leg ulcers, pulmonary hypertension, end stage renal or liver disease, progressive sickle cell retinopathy, priapism)	A full risk- benefit assessment should be carried out and each case should be considered on its own merits. Consider target HBS ≤30% and post Hb level >9 gm/dl

9. Indications for RCE during acute events:

Indication	Target
ACS with hypoxemia	Target HBS ≤30% and post Hb level >9 gm/dl
Preparation for elective surgery Low risk surgery (e.g., adenoidectomy, dental procedures) Medium-risk surgery (e.g., abdominal, cholecystectomy, tonsillectomy, orthopedic, joint replacement) High-risk surgery (e.g., cardiovascular, brain)	• RCE should be considered in: ⇒ Medium-/ high risk- surgeries requiring general anesthesia and lasting for one hour or more. ⇒ Very severe phenotype (e.g., stroke, recurrent ACS, or prior severe postoperative complications). • Simple top-up transfusion can be considered in low-risk surgeries however, management should be individualized according to other-patient related risks and disease severity. • Simple transfusion in patients with Hb < 9 g/dL and RCE for patients with Hb > 9 to 10 g/dL. • Post-transfusion Hb should not exceed 11 g/dL.
Acute hepatic sequestration	Target HBS ≤30% and post Hb level >9 gm/dl
Acute multi-organ failure	Target HBS ≤30% and post Hb level >9 gm/dl
Progressive intrahepatic cholestasis	Target HBS ≤30% and post Hb level >9 gm/dl
Severe sepsis	Target HBS ≤30% and post Hb level >9 gm/dl

10. Procedure:

10.1 Call the apheresis unit (EXT. 7129) to book the patient for RCE.

10.2 Enter request for central line insertion in ALSHIFA under Procedure and contact intervention radiology (EXT. 9407) and book the appointment.

10.3 RCE procedure:

10.3.1 Assemble the following supplies:

- Spectra Optia Apheresis System
- Exchange set with standard filler
- 0.9% sodium chloride injection USP
- ACD-A (Anticoagulant Citrate Dextrose Solution A)

10.3.2 If using peripheral access:

- Both return and inlet sites require a needle of gauge 16 or 18 to accommodate the procedure pump flow rate
- Supplies for preparing the venipuncture site.
- Blood pressure cuff

10.3.3 If using a vascular access device:

- Consent for central venous catheter of 10 Fr (12 or 15 cm) insertion must be taken from the patient.
- Connect to the device according to your standard operating procedures.

10.3.4 Touch *Select Procedure*. The procedure selection screen appears.

- Select the procedure you want to perform.
- Follow the instructions below to select the procedure options.
- When you are finished selecting the options, touch ***Confirm.***
- Follow the instructions on the screen to resume the procedure.
- Touch ***Confirm***. Once the system finishes loading the procedure software. a screen appears that displays the filler and the tubing set required for the procedure.

10.3.5 Loading the Tubing Set (RCE set)

- Verify that the tubing set has not expired by checking the expiration date on the cover of the package. Document the lot number of the tubing set to use for future reference.
- You may load the tubing set up to 24 hours before the procedure if you do not lower the cassette or prime the set. Once you lower the cassette or prime the set, you must use the set during the same work shift.
- After you load the set, the system performs a test to ensure that you have loaded the correct set for the procedure.
- Touch ***Prepare Tubing Set***. The screen appears instructing you to prepare the set.
- Put the tubing set package on top of the centrifuge cover with the package label upright and facing you and remove the cover from the package.
- Take the line holder out of the package. The line holder holds the replace line (white clamps), the AC line (orange), the saline line (green), and the extra remove line (yellow clamp). The remove bag with the attached remove line is located inside the line holder.
- Take out the remove bag and the extra remove line. Put the remove bag with attached lines on the right side of the centrifuge cover.
- Take out the replace line and put the line over the right side of the front panel.
- Take out the AC line (orange) and the saline line (green) and hang the lines over the left side of the front panel.
- Set the empty line holder aside.
- Take the vent bag out of the package and hang the bag on the left side of the IV pole.
- Take out the coiled inlet line (red clamps) and remove the paper tape from the coil. Hang the inlet connection on the left end of the IV pole. Repeat this step with the return line (blue clamps).
- Hang the remove bag on the IV pole.

- Take the cassette out of the package and put the bottom of the cassette into the bottom edge of the cassette tray.
- Ensure that there is nothing lodged behind the cassette or the tray that could interfere with the loading.
- Press on the top corners of the cassette to snap the cassette into the tray.
- Take the channel out of the package and put it on top of the centrifuge cover.
- Set the empty package aside.
- Open the centrifuge door. Locate the pin on the filler latch. Raise the latch by pushing the pin toward the center of the filler while pulling the latch up.
- Turn the centrifuge so that the loading port faces you.
- Extend the lines between the cassette and the channel and ensure that they are not twisted.
- Pull the channel up through the loading port and then through the opening in the center of the filler.
- Lower the filler latch and lock it in place.
- Position the lower collar in the collar holder on the filler latch so that the inlet line (pink line) is not obstructed by the other lines.
- Ensure that either the base of the pink line aligns with the space between the two screws or is adjacent to the indentation on the filler latch.
- Grasp the centrifuge loop below the lower collar and gently pull the collar down until you hear the “click” of the locking pin as it pops out and locks the collar in the collar holder. Ensure that the notch at the base of the locking pin is visible. If the collar is locked, you can see the notch.
- Starting with the connector, insert the channel into the groove in the filler, finishing with the inlet port.
- Run your finger over the groove and push down any section of the channel that is not completely inserted in the groove. The channel must sit flush with the groove.
- Insert the narrow part of the lower bearing into the lower bearing holder and the narrow part of the upper bearing into the upper bearing holder.
- Ensure that the braided section of the loop is not twisted.
- Position the upper collar below the upper collar holder and insert the line into the holder. Pull the line up to secure the upper collar in the holder.
- Spin the centrifuge clockwise to ensure that it rotates freely. Close the centrifuge door. Touch ***<u>Load.</u>*** The system lowers the cassette and performs several tests to verify that the tubing set and the system components function properly.

- Discard the empty tubing set package, according to your standard operating procedures.

10.3.6 Follow the instructions on the screen to prime the tubing set.

- The progress of the prime is displayed on the screen. After the AC line is primed, the system sounds a tone to notify you to open the inlet saline line and the return saline line for the prime to continue.
- Before the system primes the channel, it performs a test to verify that the correct filler is installed. The Exchange Set is primed in 3 to 4 minutes. When the system finishes priming the set, it sounds a tone.
- The patient data screen appears.

10.3.7 Selecting and Performing a Custom Prime

- The system prompts you to enter data for the custom prime after it primes the tubing set and before you connect the patient.
- Two methods are available for selecting a custom prime: Select the custom prime option on the options screen. The custom prime option is available to select before you connect the patient.
- Accept the system's recommendation to perform a custom prime. The system makes the recommendation based on the custom prime recommendation limit that was configured.
- If you change the patient data so that the recommendation limit is no longer exceeded, the system will display an alert that a custom prime may no longer be necessary.
- If you perform a custom prime, the system does not perform the step to divert the saline used to prime the tubing set. The saline is removed from the set during the custom prime.
- If you perform a custom prime, you need the following supplies:

  ⇒ Fluid for the custom prime
  ⇒ Empty bag
  ⇒ Filter or blood administration set (optional)
  ⇒ Appropriate connectors to connect the inlet and return lines to the fluid and the empty bag

10.3.8 Extending the custom prime

- You can extend the custom prime by increasing the volume entered for the fluid to use for the custom prime.
- When the screen appears instructing you to connect the patient lines, touch the go back button. The screen appears that instructs you to prime the inlet and return lines. Do not perform the instructions on the screen.
- Touch ***Confirm***. The screen appears that prompts you to enter the fluid data.
- If you decide not to extend the custom prime when you reach this screen, touch the Options tab. The options screen appears. Change the custom prime button to No, and touch Confirm.

- Touch the volume button and enter the sum of the original fluid volume plus the volume of the additional fluid. For example, if the volume that you first entered for the custom prime was 250 mL and you decide to extend the custom prime by 30 mL, enter 280 mL. Touch Confirm.
- Touch Start Custom Prime.
- Touch a menu button to display the corresponding tabs during Red Blood Cell Exchange (RBCX) Procedures

10.3.9 Entering and Confirming Patient and Procedure Data

- Touch the buttons on the screen to enter the following information:
  - ⇒ Sex
  - ⇒ Height
  - ⇒ Weight
  - ⇒ Hematocrit (Hct)

10.3.10 Selecting the Exchange Type

- Touch ***Exchange***. The following list of exchange types appears:
  - ⇒ Exchange
  - ⇒ Depletion/Exchange
  - ⇒ Depletion
- Select ***Exchange*** by touching the corresponding button on the screen.

10.3.11 Entering the replacement fluid data

- Touch ***Fluid Hct***. The data entry pad appears.
- Enter the average Hct (%) of the replacement fluid for the exchange (the system assumes that the portion of the replacement fluid that is not RBC contains 19% citrate).
- The system uses data you enter, including the Hct of the replacement fluid, to determine when the target Hct has been attained. It is important that you enter an accurate Hct for the replacement fluid.

10.3.12 Entering the fluid balance

- To enter the fluid balance in mL, touch ***Volume*** and use the data entry pad to enter a negative or positive volume. The system calculates and displays the corresponding percentage.
- To enter the fluid balance as a percentage, touch ***Percent*** and use the data entry pad to enter a percentage. The system calculates and displays the corresponding volume.
- Touch ***Confirm*** to save the fluid data.
- The run values screen appears.

10.3.13 Entering and Confirming Run Values

- Enter the run targets listed below, according to the type of RBCX procedure you selected:
  - ⇒ Target Hct (%)
  - ⇒ One of the following targets:
  - ⇒ Replaced: Exchange (mL)
  - ⇒ FCR (%)
- Perform the following steps to enter and confirm the run targets and run values:

- Touch the button on the screen that corresponds to the target value you want to enter. The data entry pad appears.
- Enter the target value.
- Review the run values that appear on the screen and confirm that they are correct. A black frame appears around the button of the primary run target. If you change a value, the color of the value on the button changes from white to yellow. Values that were affected by a change appear with a yellow arrow. The arrow points up or down to indicate an increase or decrease in the value because of the change. Only the color of the last value you changed remains yellow. The values you previously changed revert to white.
- When you are finished entering and confirming the run targets and the run values, touch ***Confirm*.**

Table below contains descriptions of the run targets that appear on the run values screens.

Run Target	Description
FCR (%)	Fraction of cells remaining. Target percentage of starting defective RBC that will remain in the patient's blood at the end of the procedure. (Target defective RBC (%)/starting defective RBC (%) = FCR.) After you enter an FCR for an exchange procedure, the system calculates and displays the volume of replacement fluid that is required to complete the procedure. **Note:** You may enter the starting and target defective RBC instead of the FCR, and the system will calculate and display the FCR. To do this, touch the button for FCR on the screen and then touch **FCR** on the data entry pad. The blood drop icon ( ) appears on the data entry pad to allow you to enter the values for defective RBC.
• Starting defective RBC (%)	Percentage of defective RBC in the patient's blood at the start of the procedure
• Target defective RBC (%)	Percentage of defective RBC in the patient's blood at the end of the procedure
Minimum Hct (%)	Lowest patient Hct during the procedure
Target Hct (%)	Desired patient Hct at the end of the procedure
Replaced (mL): Exchange	Volume of replacement fluid that is required to complete the exchange procedure. If you enter a volume here, the system calculates and displays the FCR.

10.3.14 Priming the Inlet Line, the Return Line, and the Replace Line

- Follow the instructions on the screen to prime the inlet line and the return line.
- Touch ***Confirm.***
- Follow the instructions on the screen to spike the replacement fluid container, to prime the replace line, and to put the replace line into the replacement fluid detector.
- Touch ***Continue.***

- The screen appears with instructions for connecting the patient.
- If you are using a blood warmer on the replace line, follow the instructions below to connect and prime the blood warmer tubing set and to prime the replace line.
- Load the blood warmer tubing set onto the blood warmer. Be sure to leave enough tubing on each end of the set to allow connection to the replace line.
- Clamp one end of the blood warmer tubing set.
- Clamp both replace lines.
- Unscrew the luer connector on the replace line.
- Connect the female end of the luer connector on the blood warmer tubing set to the male end of the luer connector on the replace line.
- Spike the replacement fluid container with the replace line and squeeze the drip chamber.
- Unclamp both replace lines to prime the lines and fill the drip chamber on the second replace line.
- Reclamp the second replace line.
- Unclamp the blood warmer tubing set and prime the tubing set.
- Reclamp the blood warmer tubing set.
- Connect the male end of the luer connector on the blood warmer tubing set to the female end of the luer connector on the replace line.
- Put the replace line into the replacement fluid detector so that the luer connector is below the fluid detector and the manifold is above the fluid detector. The luer connector must be positioned below the replacement fluid detector.
- Ensure that you put the Luer connector below the replacement fluid detector. If the blood warmer tubing set is not positioned below the fluid detector and the tubing fills with air during the procedure, you must disconnect the blood warmer tubing set at the luer connector and manually reprime the set. If you do not reprime the set, the vent bag could overfill with air.
- Unclamp the blood warmer tubing set.
- Touch ***<u>Continue.</u>***
- When connecting a blood warmer tubing set to the return line, ensure that the tubing connection is tight. Put the luer connection no higher than 20 in (50 cm) above the return access to prevent the possibility of air entering the tubing.
- If you are using a blood warmer on the return line, ensure that you completely prime the blood warmer tubing set to remove all the air in the set before you connect the patient.

10.3.15 Connecting the Patient and Starting the Run

- Before connecting the patient, check the inlet and return lines for air. If air is present in the lines, remove the air before connecting the patient.
- Follow the instructions on the screen to connect the patient and start the run.

- Touch ***Start Run***. The system begins drawing the patient's blood into the tubing set. The main run screen appears, and the system diverts approximately 40 mL of saline that was used to prime the tubing set into the remove bag (If you performed a custom prime, the system does not perform the step to divert the saline. The saline was removed from the set during the custom prime).

10.3.16 Monitoring the Run

- The system displays information on the screens to help you monitor the progress of the run.
- To access the screens, touch the ***Run menu*** button and then the tab for the screen you want to view.
- Be sure to verify that the volumes displayed on the screens are consistent with the actual volumes in the fluid containers and in the bags.
- The ***Main Run*** screen is the default run screen and displays or provides access to the information you most likely need to monitor the run.
- If you want to return to the main run screen after viewing a different screen, touch the ***Go Back*** button, the ***Active Menu Button***, or the ***Tab for the Current Screen***. There is no tab for the main run screen.
- To access the exchange status screen, touch the ***Exchange Status*** tab.
- Use this screen to view information about the status of the run, including the patient's fluid balance, the fluids administered, the patient Hct, and the volumes exchanged and replaced during the run.
- The type of replacement fluid currently in use appears on the button on this screen. Touch this button to change the type of replacement fluid and then follow the instructions on the screens.
- The system uses the patient Hct and the replacement fluid Hct that was entered before the start of the run to predict the current patient Hct displayed on the screen. The system does not consider RBC that were released from the spleen when predicting this value.

10.3.17 Managing Citrate Toxicity

If the patient experiences citrate toxicity, perform the following steps:

- Touch the ***Pause*** button to pause the pumps. An alarm occurs indicating that the procedure was paused, and the pumps were stopped.
- Notify the attending physician of the patient's condition according to your facility's standard operating procedures.
- Touch the ***Go Back*** button to go to the run values screen and decrease the AC infusion rate as desired.
- Touch the active alarm button to return to the alarm screen, and then touch ***Continue.*** The run values screen appears.
- Review the run values and confirm that they are correct.
- Touch ***Confirm*** to resume the procedure.

- If decreasing the AC infusion rate does not alleviate the symptoms, touch the ***Pause*** button to pause the pumps again, and notify the attending physician.

10.3.18 Giving a Fluid Bolus

- Touch the ***Run Menu*** button. The run tabs appear.
- Touch the ***Bolus*** tab. The fluid bolus screen appears.
- Follow the instructions on the screen to spike the saline container on the replace line. ***Confirm*** that the replace line is not clamped.
- Touch ***Volume*** and enter the volume (mL) of the bolus.
- Touch ***Flow Rate*** and enter the flow rate (mL/min) for the bolus. If the fluid you use for the bolus is not saline, enter a flow rate that considers the percentage of citrate in the fluid and that is in accordance with your local transfusion practices.
- To begin the bolus, touch ***Start Bolus.*** The system delivers the bolus. The screen displays the progress of the bolus.
- To cancel the bolus before delivery is complete, touch ***Cancel Bolus***.
- When the bolus is complete, the system pauses. Touch ***Continue*** to resume the procedure.
- Be aware that the patient's fluid balance displayed on the exchange status screen and on the procedure summary screen does not include the volume of any bolus that was given to the patient during the procedure.
- The system reports the total bolus volume in separate fields on the screens.

10.3.19 Optimizing the Run

- It is essential to enter accurate data when performing an RBCX procedure. Small errors in the data entered can result in large deviations in the expected outcome. The following patient data and procedure data can affect the procedure outcome:
  - ⇒ Patient Hct
  - ⇒ Patient TBV
  - ⇒ Percentage of starting defective RBC
  - ⇒ Replacement fluid Hct
  - ⇒ Replacement fluid volume
  - ⇒ Patient fluid balance
- If the patient's laboratory values are not within the expected range after the procedure is completed, review the data that was entered and verify that it was accurate.
- To review the data on the report screen. Touch the ***Data*** menu button. The data tabs appear.
- Touch the ***Report*** tab. The screen appears with a list of reports. The reports are identified by the procedure date, start time, procedure type, and the patient's TBV. The report for a procedure that was just completed is labeled ***Current.***

- Touch the button that corresponds to the report you want to view. The report appears on the screen.

10.3.20 Updating the Entered Patient Hct

- Updating the entered hematocrit could be necessary under the following circumstances:
  - ⇒ The patient's condition indicates a need to draw a blood sample and compare the laboratory value to what was entered before the run was started.
  - ⇒ The AIM system detects that the RBC interface is too close to the top of the channel, and an alarm occurs.
- If you need to update the entered Hct during the run, If the pumps are not paused, touch the ***Pause*** button to pause the pumps. An alarm occurs because the procedure was paused.
- Measure the patient's Hct according to your standard operating procedures.
- Touch the ***Run*** menu button. The run tabs appear.
  Touch the ***Exchange Status*** tab. The exchange status screen appears.
- Touch the button for the patient Hct. The data entry pad appears.
- Enter the current patient Hct. A screen appears to confirm the change to the Hct.
- Follow the instructions on the screen to verify and confirm the change. The run values screen appears.
- Review the run values.
- Touch ***Confirm.***
- Return to the alarm screen and touch ***Continue*** to restart the pumps and resume the procedure.

References:

1. Connelly-Smith et al. Guidelines on the use of Therapeutic Apheresis in Clinical Practice – Evidence-Based Approach from the Writing Committee of the American Society for Apheresis: The Ninth Special Issue. J Clin Apher. 2023; 38:77–278. DOI: 10.1002/jca.22043
2. Bernard A et al. Guidelines on red cell transfusion in sickle cell disease. Part I: principles and laboratory aspects British Journal of Haematology, 2017, 176, 179–191 doi: 10.1111/bjh.14346
3. Bernard A at al. Guidelines on red cell transfusion in sickle cell disease Part II: indications for transfusion British Journal of Haematology, 2017, 176, 192–209 doi: 10.1111/bjh.14383
4. Chou S et al. American Society of Hematology 2020 guidelines for sickle cell disease: transfusion support January 2020 Vol 4, Number. 2 DOI 10.1182/bloodadvances.2019001143.
5. *Swerdlow P.* Red Cell Exchange in Sickle Cell Disease. *ASH education book 2006*
6. Spectra Optia® Apheresis System. Operator's Manual